

Digit Property Checks in C

Algorithms Test Cases Program and Execution

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1 Problem and result

This solution reads a positive integer and uses a switch-case menu to determine whether all of its digits are even or whether all of its digits are prime. The program validates the number, handles invalid menu choices, and repeats until Exit is selected. It compiled without warnings and passed all prepared tests.

2 Algorithms

Part	Algorithm
Main program	1. Read number. 2. While number <= 0, print an error and read again. 3. In a do-while loop, show choices 1 to 3 and read choice. 4. Use switch-case to call the selected check. 5. Print the result. 6. Repeat until choice 3.
All digits even	1. While number != 0, extract digit = number % 10. 2. If digit % 2 != 0, return FALSE immediately. 3. Remove the digit using number /= 10. 4. If the loop finishes, return TRUE.
Prime digit	Return TRUE only when the digit is 2, 3, 5, or 7. These are the only one-digit prime numbers.
All digits prime	1. While number != 0, extract digit = number % 10. 2. Call is_prime_digit. 3. If it returns FALSE, return FALSE immediately. 4. Remove the digit. 5. If the loop finishes, return TRUE.

3 Test case table

Category	Input	Expected output	Observed	Status
Invalid number	-5 then 0 then 2357	Reject -5 and 0 then accept 2357	As expected	Pass
Invalid choice	number 24680 choice 4	Invalid choice message	As expected	Pass
All even valid	24680 choice 1	All digits are even	All digits are even	Pass
Even negative case	12345 choice 1	All digits are not even	As expected	Pass
Even boundary digit	2 choice 1	All digits are even	As expected	Pass
Even includes zero	20 choice 1	All digits are even	As expected	Pass
Odd single digit	7 choice 1	All digits are not even	As expected	Pass
All prime valid	2357 choice 2	All digits are prime	All digits are prime	Pass
Prime negative case	2379 choice 2	All digits are not prime	All digits are not prime	Pass
Prime repeated	22222 choice 2	All digits are prime	As expected	Pass
Prime includes zero	20 choice 2	All digits are not prime	As expected	Pass
One is not prime	1 choice 2	All digits are not prime	As expected	Pass

4 C program

```
#include <stdio.h>
```

```

#define TRUE 1
#define FALSE 0

int all_digits_even(int number)
{
    int digit;

    while (number != 0) {
        digit = number % 10;
        if (digit % 2 != 0)
            return FALSE;
        number /= 10;
    }
    return TRUE;
}

int is_prime_digit(int digit)
{
    return digit == 2 || digit == 3 || digit == 5 || digit == 7;
}

int all_digits_prime(int number)
{
    int digit;

    while (number != 0) {
        digit = number % 10;
        if (!is_prime_digit(digit))
            return FALSE;
        number /= 10;
    }
    return TRUE;
}

int main(void)
{
    int number, choice;

    do {
        printf("Enter a positive integer: ");
        if (scanf("%d", &number) != 1) {
            printf("Invalid input. Enter an integer.\n");
            return 1;
        }
        if (number <= 0)
            printf("Invalid input. The number must be positive.\n");
    } while (number <= 0);

    do {
        printf("\n1. Check whether all digits are even\n");
        printf("2. Check whether all digits are prime\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");

        if (scanf("%d", &choice) != 1) {
            printf("Invalid menu input.\n");
            return 1;
        }

        switch (choice) {
            case 1:
                if (all_digits_even(number))
                    printf("All digits are even.\n");
                else
                    printf("All digits are not even.\n");
                break;

            case 2:
                if (all_digits_prime(number))
                    printf("All digits are prime.\n");
                else
                    printf("All digits are not prime.\n");
                break;

            case 3:
                printf("Program terminated.\n");
                break;
        }
    } while (choice != 3);
}

```

```

        default:
            printf("Invalid choice. Select 1, 2, or 3.\n");
        }
    } while (choice != 3);

    return 0;
}

```

5 Compilation

```
gcc -std=c11 -Wall -Wextra -pedantic digit_checks.c -o digit_checks
```

Result: compilation completed successfully with no warnings or errors.

6 Execution and testing

```

(kali㉿kali)-[~/demo1]
$ gcc -std=c11 -Wall -Wextra -pedantic digit_checks.c -o digit_checks

(kali㉿kali)-[~/demo1]
$ ./digit_checks
Enter a positive integer: 24680

1. Check whether all digits are even
2. Check whether all digits are prime
3. Exit
Enter your choice: 1
All digits are even.

1. Check whether all digits are even
2. Check whether all digits are prime
3. Exit
Enter your choice: 2
All digits are not prime.

1. Check whether all digits are even
2. Check whether all digits are prime
3. Exit
Enter your choice: 4
Invalid choice. Select 1, 2, or 3.

1. Check whether all digits are even
2. Check whether all digits are prime
3. Exit
Enter your choice: 3
Program terminated.

```

Figure 1 Compilation and execution in Kali Linux

Function test result

All 12 function tests passed.

Representative run 1

```

Enter a positive integer: 24680
1. Check whether all digits are even
2. Check whether all digits are prime
3. Exit
Enter your choice: 1
All digits are even.
Enter your choice: 2
All digits are not prime.
Enter your choice: 4
Invalid choice. Select 1, 2, or 3.
Enter your choice: 3
Program terminated.

```

Representative run 2

```
Enter a positive integer: -5
Invalid input. The number must be positive.
Enter a positive integer: 0
Invalid input. The number must be positive.
Enter a positive integer: 2357
Enter your choice: 2
All digits are prime.
Enter your choice: 3
Program terminated.
```

7 Conclusion

The menu-driven program correctly checks whether every digit is even and whether every digit is prime. All valid, invalid, and boundary use cases produced their expected results.